

Natural Selection Study Guide

This study guide provides examples for the topics that you might see on the exam over natural selection. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered. *Macro = Δ across diff species over long period of time (new species form)*

Micro = Δ within the same species over short period. (new traits form)

1. Which of the scenarios below would be considered macroevolution?
 - a. A population of antelope that lives at high elevation for several generations experiences an increase in the frequency of an allele that contributes to increased oxygen levels in the bloodstream
 - ☒ b. A population of songbirds colonizes a remote island system and over several million years diversifies into a dozen different species specialized on different food sources
 - ☒ c. A population of mosquitoes is exposed to an insecticide and over a few generations, develops partial resistance to the insecticide
 - ☒ d. In a population of lizards, two body colors occur, black and light tan. In response to climate warming in recent decades, the light tan coloration has increased in frequency in the population

2. In class, we discussed the frequency-dependent selection ^(FDS) example of body color in female dragonflies. In this species, males prefer to mate with the females with the andromorph body color when they are less than half of the population; however, when andromorph body color is present in more than half the female population, males prefer to mate with females of the gynomorph body color. What factor(s) drive(s) body color selection in this system? *Select all that apply*

- ☒ a. Rarity
- b. Camouflage
- ☒ c. Male mate preference
- d. Female mate preference
- e. Exposure to predators

*FDS: Fitness depends on rarity of a trait.
Also its his preference!*

3. After tracking a population of birds for several years, you plot the relationship between wing length and fitness, as shown in the figure below. This figure represents [blank1] selection and the next generation of birds should have [blank2] wing lengths compared to the current generation.

Options for Blank 1: directional, stabilizing, disruptive, no
Options for Blank 2: larger, smaller, similar

Dir: One extreme is favored. $\nearrow \searrow$ "one extremely direction"

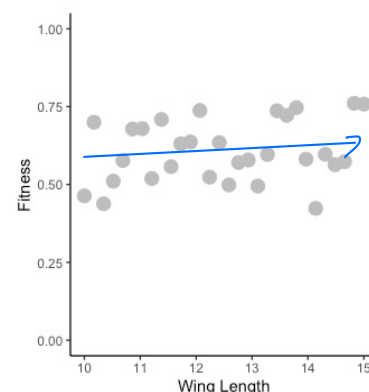
- Population shifts towards a consistently better trait
- Ex) Birds pop will shift to best beaks!

Stable: Middle is favored. $\cap$ "Stable middle"

- Middle has highest fitness; Extremes lose
- Ex) Medium birthweight babies survive best

Disruptive: Both extremes are favored $\cup$ "U are disruptive"

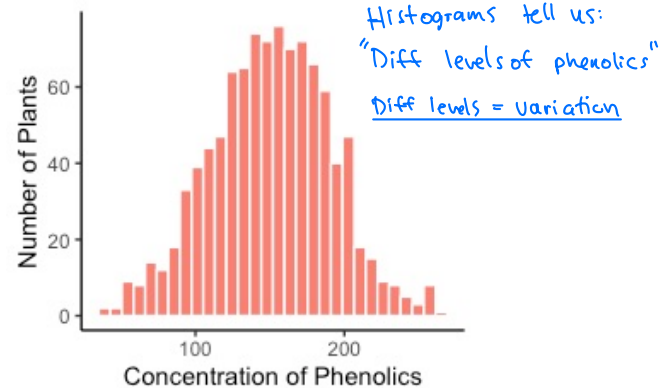
- Two "good" traits; Middle gets eliminated
- Ex) Small/Large work, but mid ones don't



Anything that keeps/adds multiple traits/alleles in a population over time (Keeping diversity alive).

4. Which of the following are examples of **balancing selection**? Select all that apply
- a. Small plants have higher fitness in summer and large plants have higher fitness in winter, resulting in a population that has both small and large plants. *diverse ✓*
 - b. When birds lay too many eggs, they can't provide for them all, but when they lay too few they are more likely to lose them all to predation, resulting in medium-sized clutches
 - c. Green morphs of an insect species have higher fitness on leaves while brown morphs have higher fitness on bark and branches. Both persist in the population. *Diverse ✓*
 - d. The establishment of a new highway causes higher fitness for birds with shorter wings because they can veer away from oncoming cars. *preference ✗*

5. Many plants produce chemical compounds that protect them from predators, either because they make the plants unpalatable (taste bad) or toxic. You hypothesize that a newly invading insect species will cause selection for higher levels of phenolics, a particular type of defensive compound. You measure plants in a meadow that has not yet been exposed to the invasive insect and plot the levels of phenolic compounds in the histogram below. Which of the necessary and sufficient conditions for natural selection does this data demonstrate?



- a. Variation = indivs differ
 - b. Heritability = traits are passed down
 - c. Differential Success = some indivs reproduce more
 - d. All of them
 - e. None of them
- "Nat select needs VHDs"*

6. You are in charge of a strawberry breeding program and are trying to use artificial selection to breed larger strawberries. Your most recent crop has an average berry size of 11.5 grams. You choose a subset of plants that make the largest berries to breed for the next generation. The average berry size for this subset is 12.8 grams. After breeding, the next generation has an average berry size of 12.2 grams.

$R = h^2 S$
 $\Delta(\text{gen1, gen2})$
 $= h^2 (\text{Selected breed} - \text{original pop})$

- Response = $\Delta \text{ gen 1 + 2}$, gen1 = 11.5, gen2 = 12.2*
- a. What is the response to selection in this system?
 $R = 12.2 - 11.5 = 0.7$
 - b. What is the heritability of this trait? *Selection differential (Selected breed - original pop) = S*
 $S = 12.8 - 11.5 = 1.3$ $h^2 = R/S = 0.7/1.3 = 0.538$

7. What is the difference between a pattern of selection and a driver of selection?

P: desc how traits change in pop (stable, dir, disrupt). D: The factor causing those changes

8. What are the necessary and sufficient conditions a population must meet in order for natural selection to occur?

"VHDs"

among the pop: Variation in traits, heritable traits, Differential success (survive longer than others)

9. How is frequency-dependent selection similar to and different from directional selection?

Both (+) and (-) FDS looks like directional over short periods. Diff: Driver for FDS is always freq of pheno vs anything else (pred camp, toxins)

10. Under what conditions would **altruism** be supported by natural selection? *(Alarm calling squirrels, Food Sharing Bats)*

Any time it increase fitness. Inclusive fitness: helping relatives reproduce. Direct Benefits: going mates, reciprocal altruism: receive help in return in the future

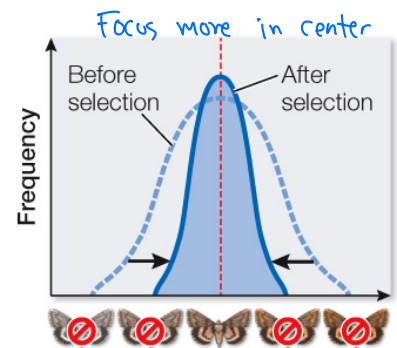
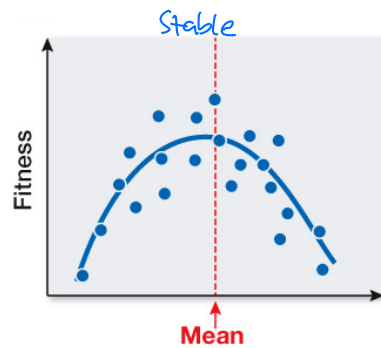
11. What types of factors constrain the process of natural selection? Phrased another way, why does natural selection not always result in organisms that are 'perfectly' adapted to their environment? *Physical: insect bodies limit by oxygen in atmos*

Evol History: NS can only mod existing traits.

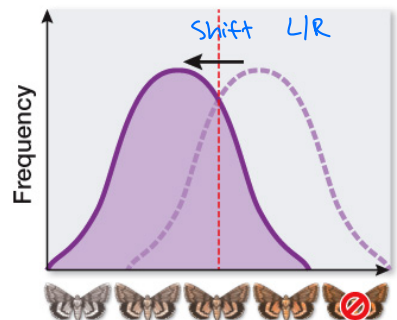
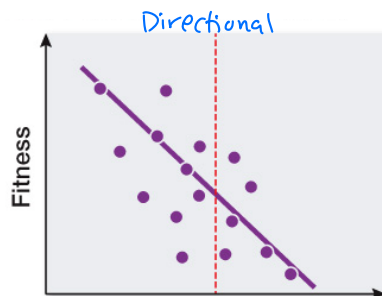
Trade offs: Improving one trait makes another worse.

12. Describe in words the relationships shown in the graphs at below. Label each pair as illustrating directional selection, stabilizing selection, or disruptive selection.

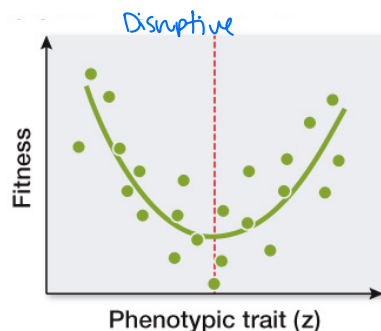
• Stable: Avg phenotype has higher fitness compared to extremes. The pop will center around the mean which reduces pheno variable (stabilize)



• Directional: Shows one extreme on the rhs. One pheno has higher fitness so the pop shifts towards the beneficial one. It can also shift to LHS as well.



• Disruptive: Both extremes have higher fitness than avg pheno which will result in a bimodal distribution. The graph forms two humps where both extremes become more common than avg.



13. Bacteria are asexual organisms, and a bacterium reproduces by duplicating its DNA and splitting into two separate cells. If a single bacterial cell that is sensitive to vancomycin (an antibiotic) is placed in an environment that contains vancomycin, it will die. Now consider another single bacterial cell, also sensitive to vancomycin, that is allowed to divide for many generations into a larger population. If this population is placed into vancomycin-containing environment, some bacteria will grow. *Diff success + Mutations!*

- Why do you see growth in the larger population (that is allowed to divide), but not with the single cell? *As bacteria divide they exp mutations. Some bacteria formed resistance while some didn't*
- Which of the populations meets the conditions for natural selection? What type of selection is occurring? *The larger colony since it displays variation in resistance and diff success in survival. We would see directional selection towards vancomycin resistance.*

14. You are studying whether harvesting large fish from a population will select for smaller body size in the fish. You create an experimental population of fish, all the same age. The average body size of your experimental population is 100 grams. You then harvest the largest fish, removing them from the population and preventing them from reproducing. The average body size of the

$$R = h^2 S$$

$$R = \Delta \text{ avg (offspring - original)}$$

$$S = \text{selected} - \text{original}$$

$$= 75 - 100 = -25$$

new mean = original + R
body size

remaining fish is 75 grams. The fish remaining are all allowed to breed. What is the predicted body size for their offspring if heritability (h^2) of body size is.....

- a. 1 $h^2 = 1$; $R = 1 \cdot -25 = -25$ $100 - 25 = 75$
- b. 0.5 • Trait is fully heritable ($=1$) so the next gen has the same avg value as the fish who reproduced.
- c. 0

$$h^2 = 1/2: R = 1/2 \cdot -25 = -12.5 \quad 100 - 12.5 = 87.5$$

- Trait is partially heritable ($=1/2$) and partially environmental.
- Harvesting of big fish does have impact in next gen but not as strong

$$h^2 = 0: R = 0 \Rightarrow 100 - 0 = 100$$

- Trait is not heritable at all and 100% environmentally controlled.
- Next gen will have the same traits as initial population.